

Formalism for the Matter Density Matrix in an ADM $3 + 1$ foliation of spacetime

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We propose to solve the problem of the statistical physics of a quantum many body fluid within the framework of general relativity working on the phase space of the particles instead of the metric. We give a formal expression for the Matter Density Matrix (MaDM) that is amenable to be evaluated through the path integral Monte Carlo method. We propose this as the simplest route to an estimation of the necessary general relativity corrections to the equilibrium and stability of a white dwarf.

Keywords: White dwarf;Jellium;Path Integral;Monte Carlo;General Relativity;Thermal Equivalence Principle

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I. INTRODUCTION

White dwarfs [1] are the dense remnants of stars, composed mainly of carbon and oxygen nuclei, which contain electrons, protons and neutrons and are supported against collapse induced by gravity by the electron degeneracy pressure due to Pauli exclusion principle. They remain white dwarfs till gravity overcomes electron degeneracy pressure, crushing electrons into protons to form neutrons through β^+ decay.

Therefore the simplest model for the interior of a white dwarf is that of *Jellium*. This is a system of pointwise electrons immersed in a uniform neutralizing background. The density of the electron gas is so high that the Fermi temperature is much higher than the temperature of the star and the electron gas can be considered degenerate. Chandrasekar [2–5] model of a white dwarf equilibrium structure didn't take into account the effects of general relativity. First Tolman [6], and later Oppenheimer and Volkoff [7] proposed a way to treat the hydrodynamic equations for the equilibrium and stability of a star within the framework of general relativity.

In order to make some progress towards a more realistic equation of state it is necessary to drop the assumptions of an ideal and perfect electron gas. This has been accomplished with various mean field theories. Since for a white dwarf the Wigner-Seitz radius is small, $0.0003 < r_s < 0.01$ the corrections due to the Coulomb interaction will be small. The main effect of the electrostatic corrections is to give smaller radii and larger central densities compared with Chandrasekhar models of the same mass. As described in §2.4 of the book [1] a first electrostatic correction arises because the positive charges are not uniformly distributed in the gas but are concentrated in individual nuclei. A more accurate equation of state is needed for the strongly correlated plasma at high temperature and high density in the classical regime or at low temperature and low density in the quantum regime. Taking linearly into account charge screening in the electron-ion plasma a better approximation can be obtained from the Debye-Hückel theory in the classical regime or the Thomas-Fermi (and Thomas-Fermi-Dirac or Hartree-Fock) theory in the quantum regime. An

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even more accurate but non linear theory of screening is offered by Vlasov theory in the classical regime or Lindhard (also known as random phase approximation) theory in the quantum regime [8]. Apart from these mean field theories, perturbative [9] or integral equation [10] approaches are also available.

More progress towards an exact solution of the model of the plasma is otherwise possible employing numerical computer experiments like path integral Monte Carlo [11] at finite non zero temperature or diffusion Monte Carlo [12, 13] at zero temperature. In order to take into account the effects of general relativity in the equation of state to be used within the Tolman-Oppenheimer-Volkoff equilibrium it is then necessary to solve a path integral Monte Carlo experiment for an electron or an electron-ion plasma within general relativity.

We can then proceed following two distinct routes, either we treat the quantum particles as living in a classical (non-quantum) spacetime or classical (non-quantum) particles living in a quantum spacetime [14]. In this letter we propose to use the former route.

This welding of *causality*, inherent in general relativity, and *casuality*, inherent in the Monte Carlo method, is expected to offer an operative approach to the problem of (quantum) statistical gravity of many body liquids [14–19]. The melt between the superposition principle of quantum physics and the non linearity of general relativity is still under construction.

II. MATTER DENSITY MATRIX

In Ref. [14] we discussed a thermal equivalence principle that should hold when dealing with a statistical physics theory of a many body fluid within the framework of general relativity: Einstein field equations $G_{\alpha\beta} = (G8\pi/c^4)T_{\alpha\beta}$ suggest that the statistical properties of *spacetime*, determined by the Einstein-Hilbert action [15–17], and the statistical properties of *matter* moving in it, determined solely by the right hand side of Einstein equations, the stress energy tensor, should be equivalent, one should be the shadow of the other. In particular when looking at the right hand side we may assume for the density matrix of the many (scalar) particle system the following expression

$$\hat{\rho} = \exp \left(- \int_{\partial\Omega} \hat{T}^{\alpha\beta} \beta_{\alpha} dS_{\beta} \right), \quad (2.1)$$

that holds in quantum statistical mechanics within special relativity [20]. In Eq. (2.1) we use Einstein convention to tacitly assume a sum over repeated indexes. We will assume that this expression continues to hold also within general relativity. We will use a hat to denote the corresponding operator, $\partial\Omega$ to denote an arbitrary spacelike hypersurface bounding the four dimensional region Ω of the curved spacetime occupied by the N particles, and β^{α} a vector such that $\sqrt{\beta_{\alpha}\beta^{\alpha}} = 1/k_B T$ with k_B Boltzmann constant and T the absolute temperature.

Within Arnowitt, Deser, and Misner (ADM) 3 + 1 foliation of space and imaginary time [17] we can evaluate the integral at the exponent of Eq. (2.1) choosing a hypersurface orthogonal to the time component so that $dS_0 = \prod_{\mu=1}^N \sqrt{{}^3g(\mathbf{r}_{\mu})} d\mathbf{r}_{\mu}$ where ${}^3g = \det\{g_{ij}\} > 0$ with g_{ij} the metric on the spatial components and $\mathbf{r}_{\mu}$ the *shifted* spatial positions of the μ particle, with $d\mathbf{r}_{\mu} = d\mathbf{x}_{\mu} + \mathbf{N}cdt_{\mu}$. Here $\vec{x}_{\mu} = (t_{\mu}, \mathbf{x}_{\mu})$ is the 4-vector event for particle μ and $\mathbf{N}$ is the *shift* in the ADM foliation

$$ds^2 = g_{\alpha\beta}dx^{\alpha}dx^{\beta} = L^2c^2dt^2 + g_{ij}(dx^i + N^icdt)(dx^j + N^jcdt),$$

with

$$\|g_{\alpha\beta}\| = \begin{pmatrix} L^2 + N^iN_i & N_i \\ N_i & g_{ij} \end{pmatrix}, \quad (2.2)$$

where L is the *lapse* and N^i the i th spatial component of the shift (see Eq. (4) in [17]).¹ We will generally use a Greek index to denote one of the 4 spacetime components and a Roman index to denote one of the 3 spatial components. A bold face symbol denotes its status as a three dimensional vector.

We may then introduce

$$\tilde{g}_{ij}(R) = g_{i_1j_1}(\mathbf{r}_1) \otimes \dots \otimes g_{i_Nj_N}(\mathbf{r}_N), \quad (2.3)$$

$$\tilde{g}(R) = \prod_{\mu=1}^N {}^3g(\mathbf{r}_{\mu}), \quad (2.4)$$

¹ For example for Schwarzschild metric outside a spherical shell the shifts are zero $N^2 = (1 - R_s/r)$ and $g_{rr} = (1 - R_s/r)^{-1}$, $g_{\theta\theta} = r^2$, $g_{\varphi\varphi} = r^2 \sin^2 \theta$ with $R_s = 2G\mathcal{M}/c^2$ the Schwarzschild radius with $\mathcal{M}$ the mass of the star inside the spherical shell, by Birkhoff theorem.

where $||\tilde{g}_{ij}||$ is a matrix made of N diagonal blocks $||g_{i_\mu j_\mu}||$ with $\mu = 1, 2, \dots, N$ and $R = \{\mathbf{r}_1, \dots, \mathbf{r}_N\}$ the shifted spatial positions of the N particles.

By the definition of the stress-energy tensor

$$\int \hat{T}^{00} dS_0 = \hat{\mathcal{H}}_N, \quad (2.5)$$

where $\hat{\mathcal{H}}_N$ is the Hamiltonian operator of the N particles *without* their gravitational potential energy. For a fluid invariant under translations and rotations² this is the only stress-energy tensor component entering the density matrix in Eq. (2.1) [20].

We then reach the following expression

$$\hat{\rho} = \exp\left(-\beta_0 \hat{\mathcal{H}}_N\right), \quad (2.6)$$

satisfying to Bloch equation

$$\frac{\partial \hat{\rho}}{\partial \beta_0} = \hat{\mathcal{H}}_N \hat{\rho}. \quad (2.7)$$

Here we choose $\beta_0 \sqrt{\tilde{g}^{00}} = 1/k_B T$ or $\beta_0(R) = 1/k_B \tilde{T}(R) = \sqrt{\tilde{g}_{00}(R)}/k_B T$ with $\tilde{T}(R) = T/\sqrt{\tilde{g}_{00}(R)}$ the Tolman-Ehrenfest temperature [21, 22]. This procedure is quite unorthodox since it assumes already a position representation for the temperature but as we will see later it correctly *includes* the gravitational potential energy. On the other hand it neglects the self-gravitation of the particles. In order to include them one should solve a self consistent field problem. Instead of treating the spacetime metric as a fixed background, the gas particles themselves act as a source of stress-energy that alters the spacetime curvature [6, 7, 23].

The coordinates, R , representation of the density matrix (2.6) can then be expanded into a path integral [11]

$$\rho(R, R'; \beta_0) = e^{-\mathcal{S}_N/\hbar} = \int \rho(R, R_1; \tau) \cdots \rho(R_{M-1}, R'; \tau) \prod_{k=1}^{M-1} d^{3N} R_k, \quad (2.8)$$

where $\mathcal{S}_N = \int_0^{\hbar\beta_0} \mathcal{H}_N dt$ is the action, $\tau(R) = \hbar\beta_0(R)/M$ is a small imaginary *timestep*, and the *primitive approximation* [11] to the high temperature, $M \gg 1$, density matrix

$$\rho(R, R'; \tau) \propto \sqrt{\tilde{g}(R)} \exp - \frac{1}{\hbar} \left[\sum_{\mu=1}^N g_{ij}(\mathbf{r}_\mu) (r_\mu^i - r_\mu'^i) (r_\mu^j - r_\mu'^j) / 4\lambda \tau(R) + \tau(R) \mathcal{U}_N(R) \right], \quad (2.9)$$

where $\mathcal{H}_N = \mathcal{K}_N + \mathcal{U}_N$ with $\mathcal{U}_N$ the potential energy due to the N particles mutual interactions and we assumed *non relativistic* $k_B T \ll mc^2$ [14] particles with a kinetic energy in a local Lorentz frame $\mathcal{K}_N = -\lambda \sum_{\mu=1}^N \nabla_\mu^2$ with $\lambda = \hbar^2/2m$. Fully relativistic particles with a dispersion relation in a local Lorentz frame, $\epsilon(\mathbf{p}) = \sqrt{(\mathbf{p}c)^2 + (mc^2)^2}$ with $\mathbf{p} = -i\hbar\nabla$, are *non-local* even in the Lorentz frame since the Taylor expansion of ϵ in $\mathbf{p}$ has ever increasing powers of $\mathbf{p}$. In order to restore locality is therefore necessary to imagine for each particle the presence of its *antiparticle* [24] with negative energy $\bar{\epsilon}(\mathbf{p}) = -\sqrt{(\mathbf{p}c)^2 + (mc^2)^2}$. In the action in Eq. (2.9) we recover the classical kinetic energy $\sum_{\mu=1}^N \frac{1}{2} m (ds/\tau)^2$ where ds is the proper distance and τ is the Tolman-Ehrenfest timestep. Attempts have also been tried in the fully relativistic case like the use of Fock ‘fifth parameter’ by Feynmann which nonetheless, despite its intuitive and creative nature, brings about some new physical interpretations not yet tested experimentally like the Hawking black hole evaporation as described in chapter 25 of the book [25].

In Eq. (2.8), at each timeslice $t_k = k\tau$, for $k = 1, 2, \dots, M-1$, we consider the section of spacetime at constant imaginary time and construct the representation of the high temperature density matrix on that spatial foil. Eq. (2.9) reduces to the usual expression in Euclidean space [11]. It has been found that one may need corrections due to the curvature of space and to the van Vleck determinant [25, 26]. In the most general case one would like to use self consistently the metric g_{ij} obtained solving Einstein field equations for the particular snapshot of the particles configuration at the given timeslice.

In the high temperature regime one recovers the *classical statistical mechanics* For $M = 1$ and $R \approx R'$

$$\rho(R, R'; \tau) \propto \sqrt{\tilde{g}(R)} \exp - \frac{1}{\hbar} [\tau(R) \mathcal{U}_N(R)], \quad (2.10)$$

² Here we mean invariant under the relative killing vector or isometry, i.e. a transformation that leaves the metric tensor unchanged.

where $\tau(R) = \hbar\beta_0(R)$.

Up to here we treated the particles as distinguishable. On the other hand the properties of a quantum many body fluid depend on the kind of quantum statistics, either Bose-Einstein or Fermi-Dirac, obeyed by its identical particles. Therefore we are rather interested in [11, 27]

$$\rho_{B,F}(R, R'; \beta) = \frac{1}{N!} \sum_{\mathcal{P}} \text{sgn}(\mathcal{P}) \rho(\mathcal{P}R, R'; \beta), \quad (2.11)$$

$$\text{sgn}(\mathcal{P}) = (\pm 1)^{\sum_{\nu=1}^N (\nu-1)C_{\nu}}, \quad (2.12)$$

where $\mathcal{P}$ is any permutation of the N particles such that $\mathcal{P}R = (\mathbf{r}_{\mathcal{P}1}, \mathbf{r}_{\mathcal{P}2}, \dots, \mathbf{r}_{\mathcal{P}N})$, with sign $\text{sgn}(\mathcal{P})$. For bosons $\text{sgn}(\mathcal{P}) = +1$. For fermions an even (odd) permutation has $\text{sgn}(\mathcal{P}) = +1(-1)$ and an even (odd) number of exchanges of a pair of particles. Any permutations can be broken into cycles $\mathcal{P} = \{C_{\nu}\}$ where C_{ν} is the number of cycles of length ν in $\mathcal{P}$. We may then introduce a deBroglie thermal wavelength

$$\Lambda(R) = \sqrt{2\lambda\beta_0(R)}, \quad (2.13)$$

dependent on the point in the $3N$ manifold. When the size, $\Lambda(R)$, of the many body path $R_1, R_2, \dots, R_M$ equals the interparticle spacing, $n^{-1/3}$ with n the proper particle number density, the permutations will have an effect on the measure of diagonal observables. This determines a degeneracy temperature

$$T_D = \frac{2\lambda n^{2/3}}{k_B}. \quad (2.14)$$

We may then introduce an adimensional parameter $\Theta = T/T_D$ such that for $\Theta \ll 1$ quantum statistics will be very important.

This expression correctly reduces to the density matrix of the classical, $\Theta \gg 1$, system of non relativistic, $p/mc \ll 1$, particles in weak gravity, $\Phi/c^2 \ll 1$ with $\Phi(r) = G\mathcal{M}/r$ with $\mathcal{M}$ the mass of the star. In fact $\sqrt{\tilde{g}_{00}(R)} \approx \prod_{\mu=1}^N [1 - \Phi(r_{\mu})] \approx 1 - \sum_{\mu=1}^N \Phi(r_{\mu})$. So that

$$\hbar\tau(R)\mathcal{H}_N = \frac{1}{k_B T M} \left(\sum_{\mu=1}^N \frac{p_{\mu}^2}{2m} - mc^2 \sum_{\mu=1}^N \frac{G\mathcal{M}}{c^2 r_{\mu}} + \mathcal{U}_N(R) + \dots \right), \quad (2.15)$$

where we neglected terms of second order in the small quantities p/mc and $GM/c^2 r$.

We have previously proposed [28] to consider the $\sqrt{\tilde{g}(R)}$ factors as an external geometrical potential

$$W_N(R) = -\frac{1}{k_B T M} \ln \left[\sqrt{\tilde{g}(R)} \right] + \dots, \quad (2.16)$$

where the dots stand for additional corrections due to the space curvature and the van Vleck determinant [25, 26]. These contributions can more easily be taken into account by a change of coordinates which transform the Laplace-Beltrami operator into the flat Laplacian, solving Bloch equation (2.7) in the new coordinates and then reverting to the original ones [29].

The classical limit (2.10) has been used in various spatial dimensions to describe the properties of Coulomb systems. These offer some exact analytic solutions in one dimension [30, 31], in two dimensions [32], and at a special value of the coupling constant in two dimensions on various surfaces: The plane [33], the cylinder [34], the sphere [35], the pseudosphere [36], the Flamm paraboloid [37, 38].³

On the other hand, the path integral (2.8) with the primitive approximation (2.9) has been treated for example in Refs. [25, 26] and applied in various geometries to various physical systems. For the example of the sphere treated with the Monte Carlo method [39] see Refs. [19, 40, 41].

³ In $d+1$ spacetime with $d \leq 2$ the Weyl conformal tensor vanishes and the Riemann tensor is completely determined by its trace, the Ricci tensor. Contracting Einstein field equations in vacuum implies a vanishing scalar curvature, the trace of Ricci tensor. Then Einstein field equations in vacuum reduce to the vanishing of the Ricci tensor and consequently to the vanishing of the whole Riemann tensor. It is only in $d > 2$ that general relativity looks like a theory of the gravitational interaction. Nonetheless Flamm paraboloid is an interesting surface of non constant curvature where the Coulomb plasma still admits an exact analytic solution.

III. CONCLUSIONS

In order to extract the statistical mechanics properties of a many (scalar) particle fluid within general relativity, we chose to work on the right hand side of Einstein field equations, building a density matrix from the stress-energy tensor through a path integral where on each imaginary timeslice one selects a constant time foil of the $3 + 1$ ADM spacetime foliation.

This Matter Density Matrix (MaDM) should follow as a shadow the equivalent Metric Density Matrix (MeDM) constructed from the Einstein-Hilbert action, which gives a statistical mechanics of spacetime [17]. It is our impression that among the two equivalent routes to statistical physics this last one offers a mathematically much more cumbersome approach [42]. Nonetheless the use of the Markov stochastic processes, inherent to the Metropolis algorithm [43], to measure a structural or thermodynamic observable of the many body fluid should make no distinction among the two routes. In fact the Metropolis algorithm is very simple and powerful; it can be used to sample essentially any probability distribution function regardless of analytic complexity in any number of dimensions.

AUTHOR DECLARATIONS

Conflicts of interest

None declared.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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